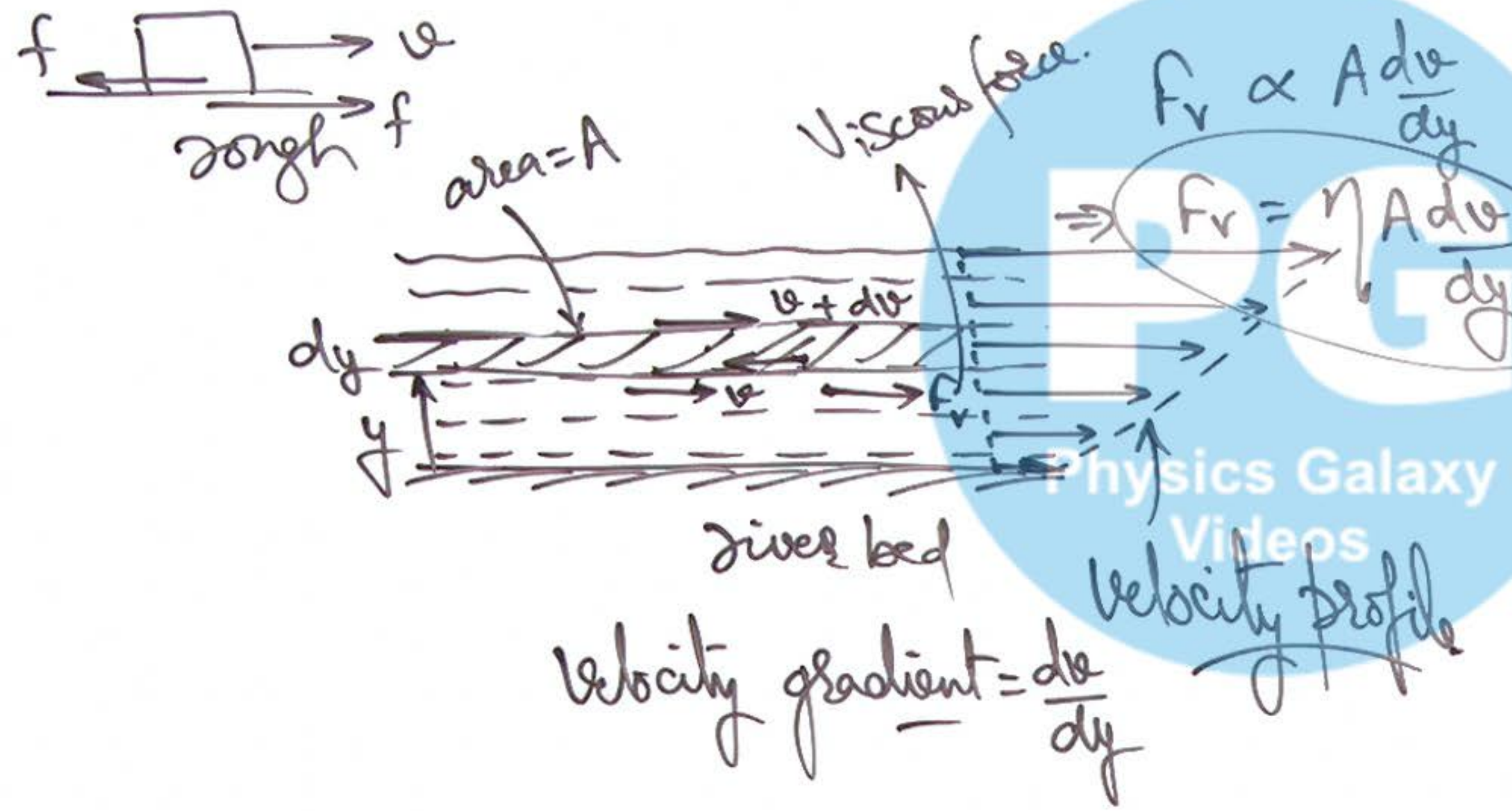


① Understanding Viscosity:



- It depends on nature of fluid (liq/Gases)

- $\eta_{\text{liq}} > \eta_{\text{gases}}$

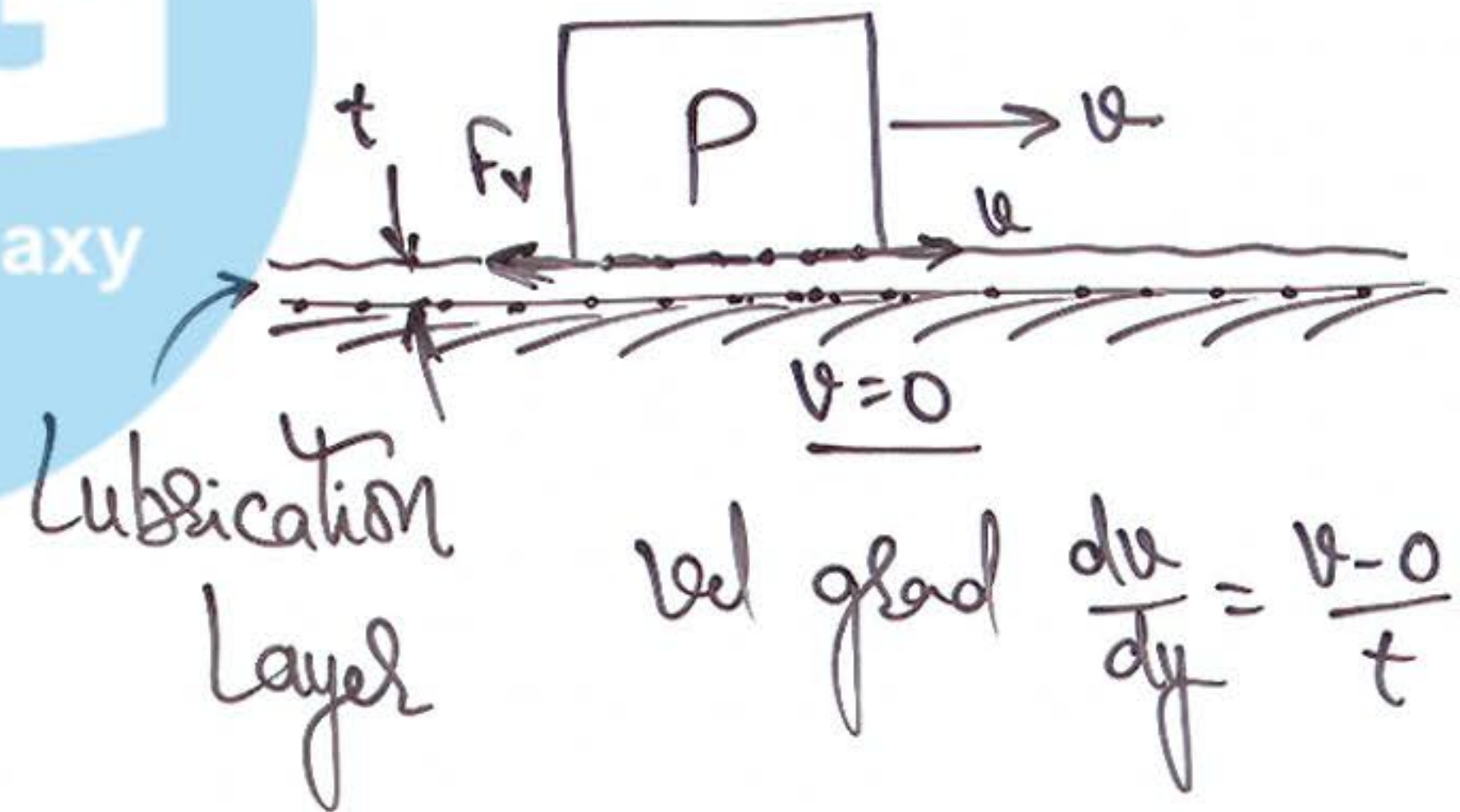
- For liquids as density $\uparrow$, $\eta \uparrow$
for gases as density $\uparrow$, $\eta \downarrow$

② Force on a Solid due to fluid layers:

Viscous force on block

$$F_v = \eta A \frac{dv}{dy} = \frac{\eta A v}{t}$$

A = bottom surface area.



Lubrication
Layer

③ Stoke's Law & Terminal Velocity:

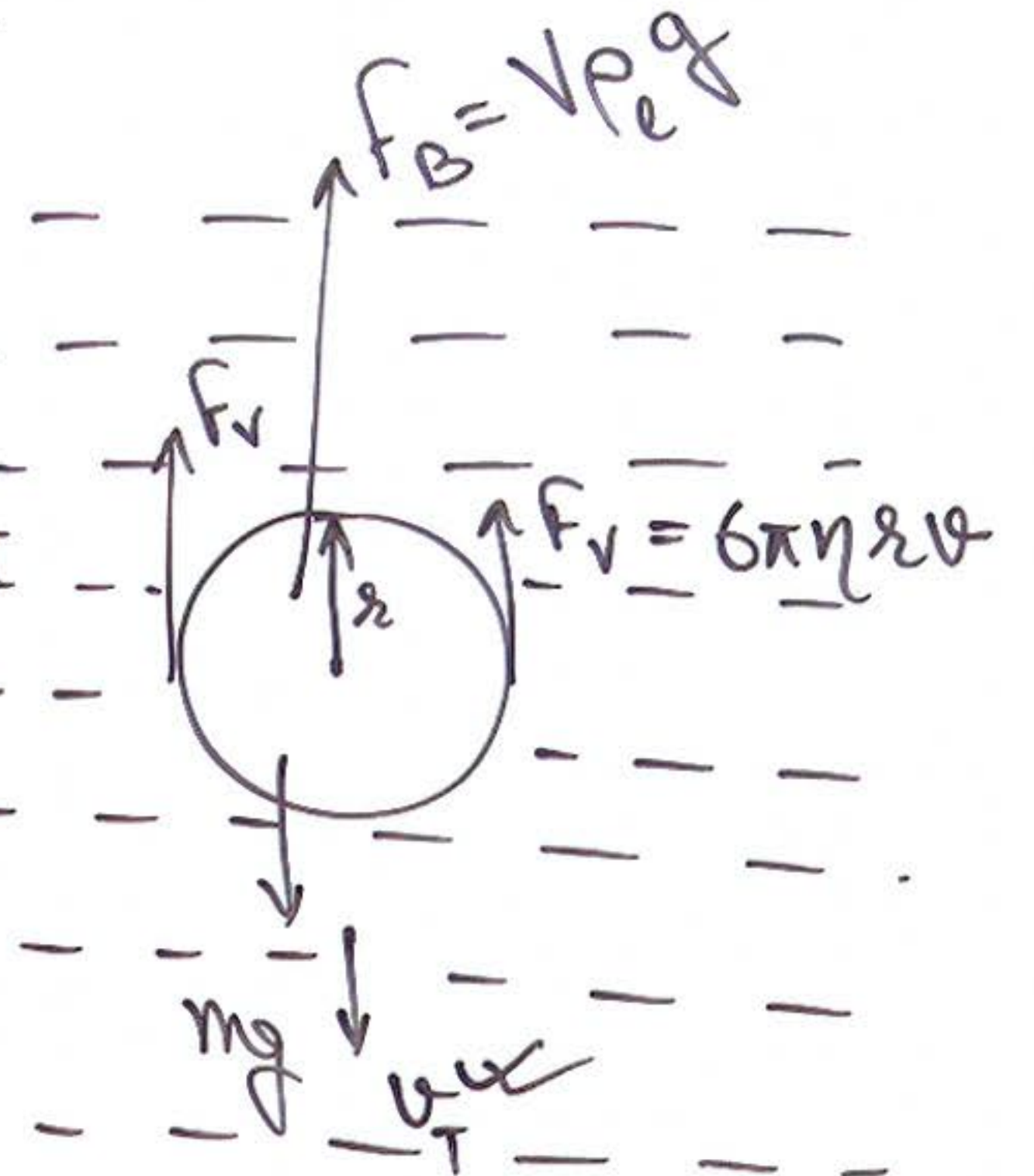
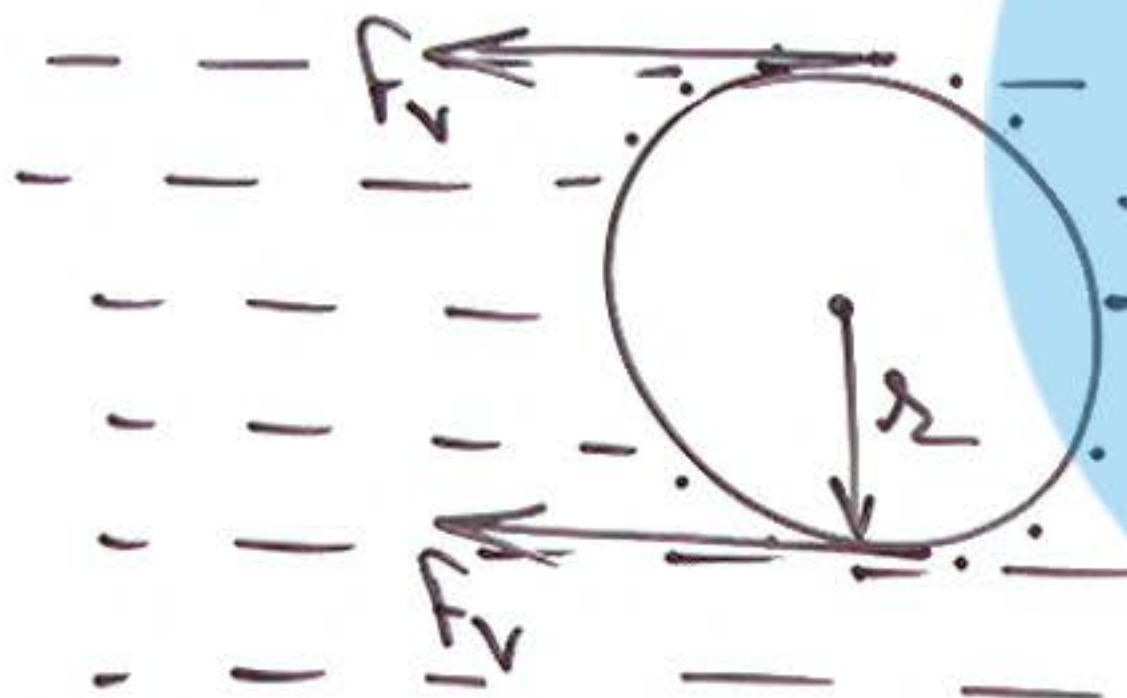
At Terminal Speed.

$$F_v = 6\pi\eta r v$$

$$V_s g = V_r g + 6\pi\eta r v_T$$

$$\frac{4}{3}\pi r^3 \rho_s g = \frac{4}{3}\pi r^3 \rho_r g + 6\pi\eta r v_T$$

$$v_T = \frac{2}{9} \frac{g r^2 (\rho_s - \rho_r)}{\eta}$$



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④ Critical Velocity:

↓
flow changes from

laminar to turbulent

if $v_f > v_c$

① $v_c \propto \eta$

② $v_c \propto \frac{1}{\rho}$

③ $v_c \propto \frac{1}{D} \left(\text{or } \frac{1}{r} \right)$

$\Rightarrow v_c \propto \frac{\eta}{\rho D}$

$v_c = N_R \left(\frac{\eta}{\rho D} \right)$

⑤ Poiseuille's Equation:

fluid flow rate $Q = Av = \pi r^2 v$

$$Q = \frac{\pi r^4}{8\eta l} (P_1 - P_2)$$

$$\Rightarrow Q = \frac{(P_1 - P_2)}{(8\eta l / \pi r^4)}$$

